

Two-Stage Water Rocket & Its Simulation Platform

Transcultural & transectoral collaboration in transdisciplinary education

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Significance

How do you teach abstract mechanics to young children across a language and sector boundary? We designed a **bilingual (Chinese–English) STEM unit** and delivered it as a **cross-regional service-learning project** — Hong Kong university students teaching primary-school children in mainland China.

Transdisciplinary physics × engineering × AI × pedagogy

Transectoral university ↔ school ↔ community

Transcultural Hong Kong ↔ Jiangxi, bilingual

Objectives

Knowledge

Explain launch principles, pressure vs. air pressure, action–reaction, acceleration and air resistance.

Ability

Practise **hypothesis** → **test** → **optimise**; variable control; data recording; engineering design.

Transdisciplinary

Couple **AI simulation** with **hands-on experiment**, transferable to other STEM topics.

In the field



Approach & Methodology

1

Learn

Rocket basics · two-stage structure · pressure

2

Build

Rocket assembly · variable control

3

Simulate

AI digital-twin parameter sweep

4

Launch

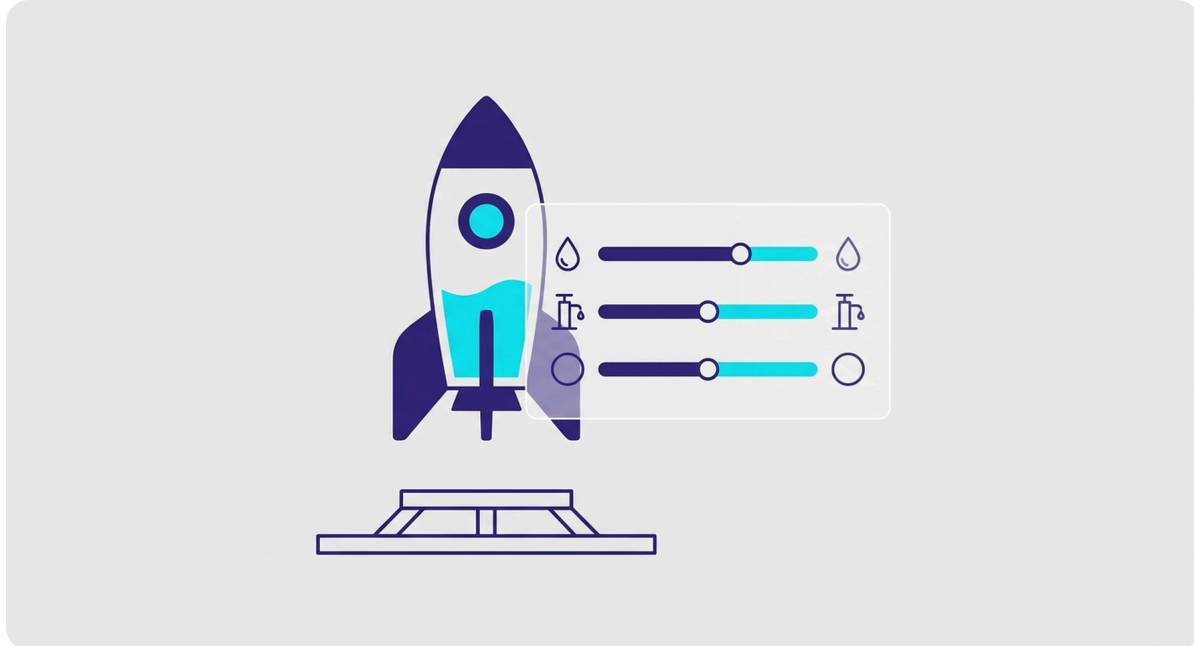
Formal launch · measure & record

5

Reflect

Simulation vs. real · optimise

A custom **AI simulation platform** acts as a **pre-launch “digital twin”**: students sweep **water volume**, **air pressure** and **nozzle diameter** to predict the optimal launch *before* the physical test — reducing cost and safety risk while grounding intuition in theory.



Materials



Hands-on

Key Findings & Implications

- Anticipated outcomes** — a target of **[X]**% of learners are expected to explain action–reaction and pressure; simulation predictions validated against **[M]** real launches.
- Stakeholder impact** — **learners** grasp core mechanics and build inquiry skills; the **school & community** gain a reusable, low-cost STEM module; we, as **facilitators**, develop transdisciplinary teamwork and cross-cultural communication.
- Next steps** — map physics one-to-one onto build/launch steps; add academic-standard scenarios; build a student innovation zone to close the loop **learn** → **practise** → **create**.

“The children connected pressure and force faster than any worksheet ever did.”

— Teacher testimony (to be confirmed)

“I learned why my rocket flew higher when I pumped in more air.”

— Student testimony (to be confirmed)



References & Acknowledgement

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Acknowledgement. We thank our advisors **Dr Simon Wang** and **Dr Nancy Guo**, the Hong Kong Baptist University Sustainable Service Education Group, the service-learning / Tin Ka Ping programme, and our partner school.



Scan to explore the full project

github.com/tesolchina/paste2026-poster

Pacific Asia Summit on Transdisciplinary Education (PASTE) 2026 · 26–27 Nov 2026 · HKBU